

Akshata Madavi

Applied AI Engineer | ML Systems | LLM Evaluation

akshatamadavi@gmail.com | linkedin.com/in/akshata-madavi | github.com/Akshata4 | +1-408-590-1556 | San Jose, CA

SUMMARY

Applied AI Engineer and MS Software Engineering student with 7 years of experience building production data systems and AI applications. Experienced in document intelligence, LLM evaluation, agentic workflows, RAG, model fine-tuning, and real-time ML deployment. Built enterprise AI systems that reduced review cycles from weeks to days.

SKILLS

- **Languages:** Python, SQL
- **Applied AI:** RAG (LangChain, LlamaIndex), agentic workflows, tool calling, document intelligence, OCR, LLM evaluation (LLM-as-judge, custom graders), LLM observability, LoRA fine-tuning
- **ML:** PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face
- **AI Infrastructure:** Vertex AI, MLflow, LangSmith, Chroma, Docker
- **Data Systems:** Spark, Kafka/KSQL, BigQuery, Airflow
- **Cloud/DevOps:** GCP, AWS (S3, DynamoDB, Redshift, EMR), Azure (Data Factory, Databricks), GitHub Actions CI/CD

EXPERIENCE

Applied AI Engineer Intern, The Mortgage Office – San Mateo, CA

Jul 2026 – Aug 2026

- Designed and productionized a Python document-intelligence service on Azure for OCR, layout parsing, and structured extraction across **100+ document types**, replacing a paid third-party API; built code-based and LLM-as-judge evaluation graders across **~1,000 test cases** that caught missed tool calls on roughly half of required-tool turns, cutting the release review cycle from weeks to days.

Senior Data Engineer, LTIMindtree – Pune, India

May 2023 – Mar 2025

- Engineered near real-time GCP/BigQuery pipelines ingesting **~10M records** from enterprise sources, building ML-ready feature tables for downstream AI/ML models at production scale.
- Redesigned a multi-source ETL service (~30 TB) with parallel Spark jobs, cutting ingestion **from 1 week to under 1 hour**; served as the primary feature store for ML workloads.

Data Engineer, AI Adventures – Pune, India

Aug 2021 – Apr 2023

- Led migration of course data to an Apache Iceberg data lake, **cutting long-term storage costs by ~50% and reducing data latency by ~95%**.
- Ran A/B testing and retention analysis on the resulting data, **driving a 20% revenue increase**.

Data Engineer, STL (Sterlite Technologies) – Pune, India

Jun 2018 – May 2021

- Built near real-time GCP pipelines migrating Oracle, SAP, and Salesforce data into BigQuery (~1 TB); applied DOE-based statistical analysis and automated anomaly detection for manufacturing process optimization.
- Consolidated multi-phase manufacturing data into a centralized warehouse and trained an XGBoost model to pinpoint the process generating the most scrap; the resulting RCA and fix **cut scrap production by ~18%**.

PROJECTS

Personal Finance Auditor – Agentic Text-to-SQL Assistant

PydanticAI, Gemini API, SQLite, sqlglot, Streamlit, Logfire, Python

- Built a multi-agent pipeline: parallel agents categorize bank-statement merchants at ingest, and a conversational agent answers plain-English spending questions through a Text-to-SQL tool-calling loop.
- Enforced a two-layer SQL safety guardrail (sqlglot parse-level validation + read-only connection) and **instrumented every agent run with Logfire traces** (LLM calls, tool calls, latency, tokens); deployed via Streamlit.

Enterprise AI Governance Platform – PII Detection & LLM Monitoring

PyTorch, HuggingFace, LoRA, LangChain, Chroma, MLflow, Docker, AWS

- Fine-tuned Qwen 0.6B with LoRA on a labeled PII dataset, tracking experiments (rank, alpha, learning rate) in MLflow to **improve detection recall from 87% to 97%** on a held-out evaluation set across email, phone, API key, and SSN entity types.
- Shipped a Chrome extension for real-time PII detection in web forms (LLM inference with regex fallback) and a LangChain + Chroma retrieval pipeline for failure-pattern analysis and compliance audit trails, containerized with Docker.

VitalWatch – Real-Time Sensor Anomaly Detection (AI-Catalyst Hackathon 2025)

PyTorch, XGBoost, Kafka/KSQL, GCP Vertex AI, BigQuery, Twilio, Python

- Trained XGBoost on multi-channel time-series sensor data, reaching **98% precision** on a held-out test set for the anomaly-detection task; deployed as a scoring service on GCP Vertex AI.
- Built a Kafka/KSQL streaming pipeline enriching flagged events via LLM with BigQuery data, delivering end-to-end alerts in **~10 seconds**.

EDUCATION

San Jose State University, MS in Software Engineering

May 2027

College of Engineering Pune (COEP), BS in Computer Engineering

May 2018